

AI-Based FPGA-Accelerated Lightweight YOLOv8n Model for Real-Time PCB Defect Detection Using the PYNQ-Z2 Board

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Abstract—Every electronic system is built on printed circuit boards (PCBs), yet manufacturing flaws including open circuits, holes and copper traces have a significant impact on PCB reliability. Conventional inspection techniques, such as automated optical inspection (AOI) and manual inspection, are expensive, time-consuming, and unable to satisfy industrial expectations in real time. This research suggests a real-time, detection of defects in PCB which is implemented in PYNQ platform that integrates hardware–software co-design to maximise performance in order to overcome these issues. A lightweight YOLOv8n deep learning model is used by the system, and PCB photos are pre-processed using resizing, filtering, and normalisation. The approach strikes a balance between resource usage and detection performance through the use of INT8 quantisation and model trimming. The FPGA fabric performs parallel convolutional operations to ensure low-latency and high-throughput inference, while the ARM Cortex-A9 processor controls system flow and result visualization. A golden board cut-off technique further enhances defect localization and highlighting. This FPGA-based architecture provides a scalable, power-efficient, and cost-effective alternative to conventional AOI systems, bridging the gap between deep learning accuracy and embedded hardware constraints for intelligent, real-time industrial inspection.

Keywords— *Printed Circuit Board (PCB), FPGA Acceleration, YOLOv8n, Defect Detection, Quantization, Real-Time Inspection, Embedded Vision, PYNQ-Z2 Board.*

I. INTRODUCTION

The advanced evolution of electronic world has grown the need for fast, real-time and automatic inspection used for Hi-Fi manufacturing facilities. Printed Circuit Boards (PCBs), as the physical and functional conventional and advanced electronic system, have to be of the utmost quality because even the slightest defects, like hairline cracks, absent vias, or random copper, can lead to serious device failures. Although accurate, traditional inspection processes such as X-ray inspection and Automated Optical Inspection (AOI) are costly, rigid, or unsuitable for small- and medium-scale

industries seeking scalable, energy-saving solutions. Object detection frameworks based on deep learning, like YOLOv3, YOLOv4-Tiny, and YOLOv5, have demonstrated good potential in defect detection, but also, they cannot be employed in embedded industrial environments because of the high-power consumption and the inability to run them on a small computer. This paper presents an FPGA-based PCB fault-detection system that combines deep learning smarts and reconfigurable hardware efficiency to overcome these challenges. The proposed system was implemented on an Xilinx PYNQ-Z2 platform running a pruned and optimized INT8-quantized Zynq-7020 SoC to deliver real-time performance within hardware resource constraints. The FPGA fabric performs convolutional computations, and the ARM processor performs preprocessing such as image resizing, filtering, and normalization. Moreover, golden board removal and post-processing with contours are combined to increase the detection accuracy of both micro- and macro-defects. Experimental verification shows the system can detect with about 90 percent accuracy and achieve significant improvements in latency and power consumption, indicating it can be implemented in industrial assembly lines. This study provides a scalable, energy-efficient, and cost-effective embedded system that connects high-performance deep learning models with limited-resource hardware platforms to detect intelligent PCBs.

II. LITERATURE SURVEY

Defects in PCB have become increasingly important in the production of modern electronics due to the need for high precision and efficiency. Conventional inspection, including hand inspection with the eye-piece, flashlights, etc., is time-consuming, laborious, and subject to error [4]. Researchers have addressed these constraints using AI-based techniques to detect PCB defects. Advanced Printed Circuit Board Defect Detection Scheme (PD-YOLOv8): The PD-YOLOv8 model improves small-target recognition by adding the Efficient Channel Attention Network (ECANet) and using C2fE for enhancing the neck architecture [1]. The design enhances

detection results of minute PCB defects, frequently overlooked by traditional designs, showing that attention mechanisms are critical in feature extraction. YOLOv10-Developed Deep Learning Architecture: Ruengrote et al. initiated the higher level version of YOLO with integrating feature extraction with higher accuracy^[2]. Their model was precise 96, had a recall of 97 and a mAP of 99, making it far outperforming the previous versions of YOLO and Faster R-CNN, and demonstrating the effectiveness of architectural additions to deep learning in PCB inspection. Lightweight model YOLOv8-PCB is used for the risks for poor adaption to combine and global features to increase their sensitivity find defects^[3]. This architecture shows that even in computationally limited environments, high detection accuracy can be achieved. YOLO-DFW with Tiny and Complex Defects: YOLO-DFW is a variant of YOLOv8 that identifies defects on PCB with high precision, by employing a C2f-Dynamic module (to extract features) and a Feature Focused Pyramid Network (to fuse features)^[4]. The method uniquely tackles issues with identifying small defects and intricate PCB forms, justifying the usefulness of multi-scale feature improvement and attention-focused feature improvement. Real-Time YOLOv5 Doing the same with YOLOv5: The FPGA-based YOLOv5 Accelerator is executed with quantization based optimization which prefers the detection efficiency^[5]. This is essential in industrial applications that need rapid and power-efficient processing. Embedded YOLOv3 Deployment on FPGA: Yu et al. developed an FPGA-based YOLOv3 model that provides an FPGA-based Deep Learning Processing Unit (DPU) to perform convolution and compresses the model through quantization^[6]. The system had a throughput of 2.44 TOPS and an accuracy of 0.789, with low power consumption required in industrial PCB inspection. FPGA Neural Network Accelerator with RepYOLOv2: RepYOLOv2 is a better version of YOLOv2 that features structure re-parameterization and low-bit 6-bit quantization, implemented on an FPGA in edge computing^[7]. This accelerator delivered a high throughput (68.53 GOP/s) and low power consumption (1.12 W), demonstrating that real-time defect detection on edge devices can be realized when hardware-aware model design is taken into account. YOLOv8n-SM to Robust PCB Defect Detection: The YOLOv8n-SM model also enhances the accuracy of detection with higher mAP0.5 and mAP0.5:0.95 and is robust to detect PCB defects across different datasets^[8]. This emphasizes the significance of custom architectural changes to enhance defect detection measures. Project implementation: In the given work, a better YOLOv8-based PCB defect detector is developed, which is based on the above studies^{[1]-[8]}. The model combines attention to extract better features, leverages lightweight architecture to make inferences quickly, and trains on a dataset of industrial PCB images labeled with defects of various types. It showed improvement in precision, recall, and mAP, which matched well with the results presented in previous studies. This implementation builds upon the experience of lightweight design^[3], multi-scale feature fusion^[4], and attention mechanisms^[1] to improve detection of small defects in a cost-effective manner.

III. MATERIALS AND METHODS

The PCB Defect Dataset applied to this project was obtained at the Human-Robot Interaction at Peking University and is also aimed at detecting objects and controlling quality in electronics production. It was initially composed of 1,386 images and 2,953 annotated defects in six labels, but subsequently was expanded to 10,668 annotated images in order to ensure a stronger model. All the images were annotated in YOLO format, and the class indices along with normalized bounding box coordinates were stored in the corresponding .txt files. The dataset has been preprocessed and made consistent and compatible with the model: all images were reduced to 640 x 640 pixels, normalized, and converted to the standard RGB format. Different data augmentation methods have been used to enhance generalization and minimize overfitting, such as horizontal and vertical flipping, random rotation, scaling, changing brightness, and adding Gaussian noise, which mimics the changes in the real-world manufacturing. The data were get divided for training and execution process. This holistic pre-processing pipeline made the YOLOv8n model a highly effective detector of defects descriptors, which allowed the accurate detection of small PCB defects, including fine spur, spurious copper, and open circuit features, thus, becoming a solid basis of high precision automated PCB inspection in production settings.

A. DATASET

The dataset types taken to use for the study is the PCB Defect Dataset, which is then obtained in Kaggle and was originally published by Human-Robot Interaction, Peking University. It is specifically aimed at detecting objects and focuses on small defects in PCB. This dataset is a total of 1,386 high-resolution images, which are annotated with 2,953 labeled instances of defects that are grouped into six specific classes of defects which includes the open and short problems and missing holes. To increase the robustness and generalization of the models, the dataset was enriched with the transformations of rotation, flipping, and scaling to obtain a larger dataset consisting of 10,668 images and the responding annotation files in YOLO format. This rich and varied data can be used in training and testing deep learning models to detect defects on PCB in a batch, a primary element of quality control in the electronics manufacturing sector. In addition to the visual defect identification, some of the future improvements of the proposed system could include the analysis of stuck-at faults at the circuit level. Stuck-at faults are generated when a node is stuck at a logic '0' or a logic '1', which is mostly due to the manufacturing defects or the inability to connect the nodes properly in digital PCBs. The stuck-at faults can be identified through logic-level testing using test vectors, boundary scan testing, or FPGA test pattern generation. The integration of structural fault testing with the proposed vision inspection system would make it possible to perform both physical and functional verification of PCBs, thus improving the reliability of the system

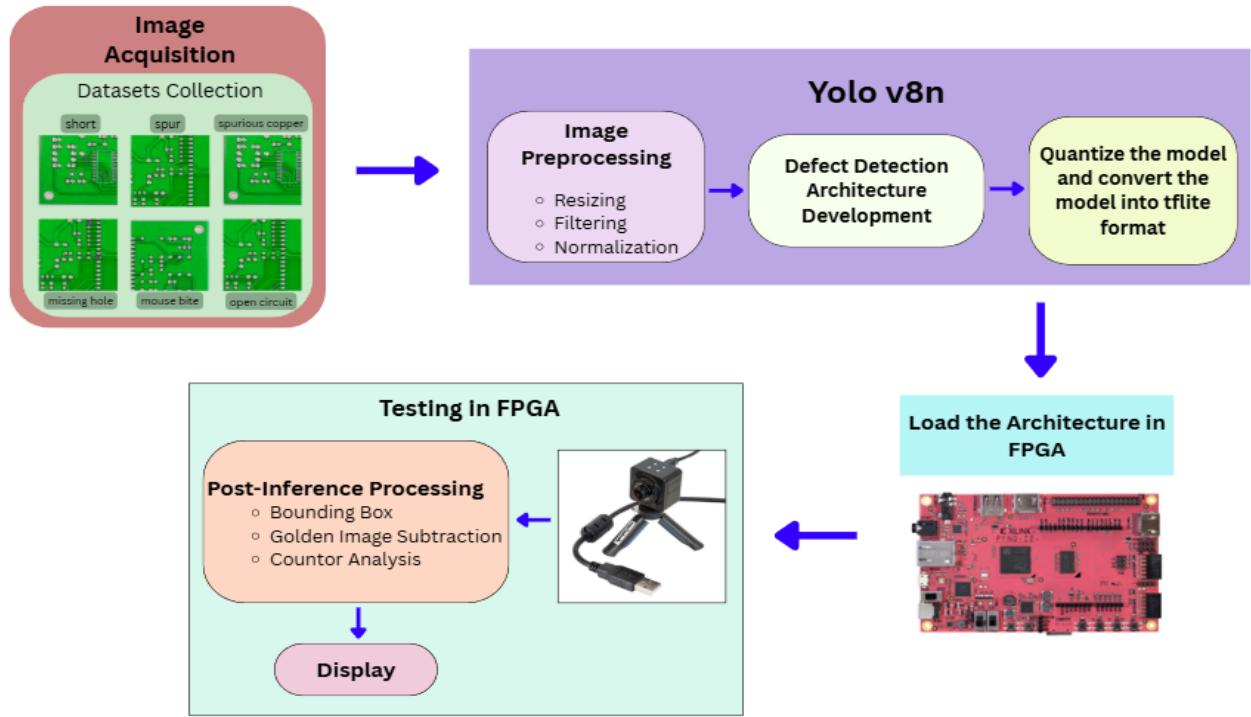


Fig. 1. Proposed Methodology

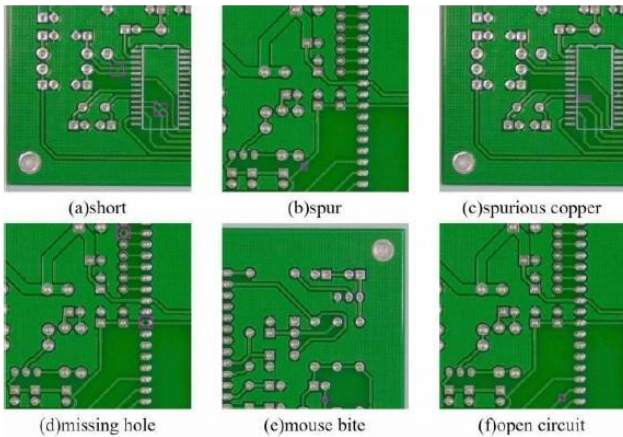


Fig. 2. Defect Types

B. YOLOV8n

YOLOv8n (You Only Look Once version 8 - nano) is a progressive real-time object detection model created by Ultralytics, which is aimed at choosing between accuracy, speed, and computational cost. Being the lightweight version of the YOLOv8 family, YOLOv8n is a highly optimized architecture designed for embedded vision systems and FPGA-based applications. The design is used to process the object bounding boxes, thereby decreasing overall cost and improves normalization across various defectiveness. Backbone Network: Learns features of input PCB images based on convolutional layers and residual connections.

1) YOLOV8n ARCHITECTURE

The YOLOv8n architecture consists of three major sections: the Backbone, Neck, and Head. The conventional model uses the C2f module (Cross Stage Partial with concatenated feature maps) that improves gradient

propagation, the usage of description and representation in various scales. Path Aggregation Network (PANet), which combines lower information and high-level information for detection of the open and short circuit errors.

Bounding boxes, confidence scores, and class probabilities are jointly predicted, enabling the system to detect and classify defects during a single forward pass with low latency. In this project, the augmented dataset is trained using the image recognition network (YOLOv8n) to discriminate and recognise errors in PCB fabrication units. The dataset is annotated and trained with 32 batches and stochastic Gradient Descent (SGD) as the optimiser for 100 epochs.

C. PYNQ BOARD IMPLEMENTATION

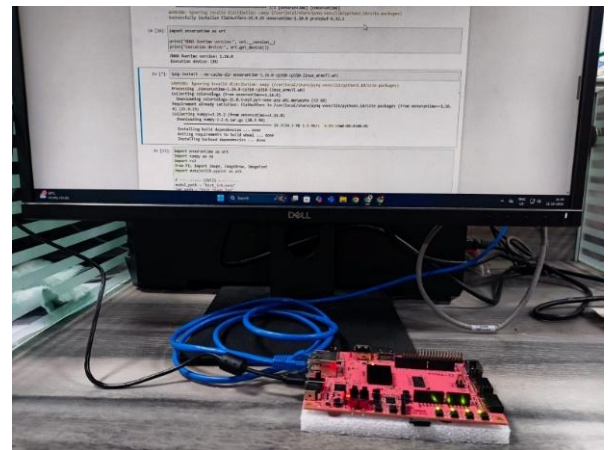


Fig. 3. PYNQ Board Implementation

The dataset of PCB Defects used in this project is from the Human-Robot Interaction at Peking University and was specifically selected for object detection and quality control in electronics manufacturing. It initially had 1,386 pictures with 2,953 annotated defect situations PCB units. All images were annotated using the YOLO format, with class indices and normalized bounding box coordinates saved in separate .txt files. A lot of preprocessing was done to make images consistent and model compatible: all images were resized to 640 x 640 pixels, normalized, and converted to the standard RGB format. In a bid to enhance generalization and reduce overfitting, types of data augmentation methods were used, including the adjustment of sizes and its orientations.. This complex process of preprocessing made sure that the YOLOv8n model was able to learn both global and local defects so that it could correctly detect fine spurs, spurious copper residues, and open circuits as well hence creating a solid basis that would allow a high precision of automated inspection in the industrial setting.

IV. RESULTS AND DISCUSSION

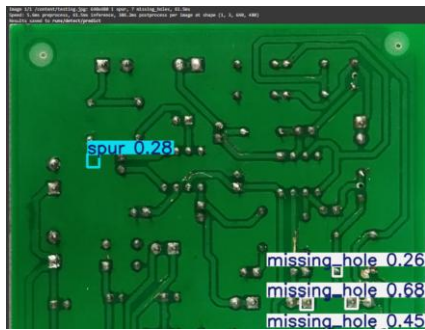


Fig. 4. Output Image – CPU

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Latency: 3.552162 seconds
ONNX output shape: (1, 10, 3549)
Valid detections: 1
Predicted classes: ['mouse_bite']

===== MODEL PERFORMANCE SUMMARY =====
Latency (First Inference): 3.5522 sec
Inference Speed (Next Run): 3.5297 sec
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000
Mean Absolute Error (MAE): 0.0000
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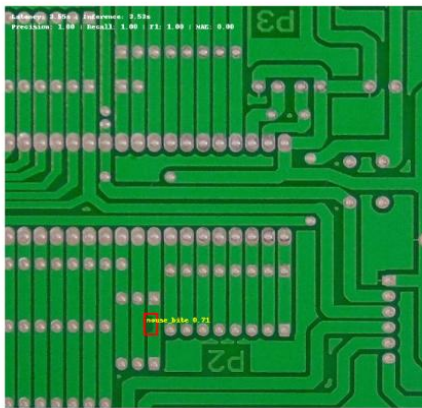


Fig. 5. Output Image-PYNQ board

Figure 4 displays the results by the YOLOv8n model running on a CPU. There are many defects visible in this image, some with confidence levels between 0.26 and 0.68. It proves the model can classify fault types in PCB tracks and pads since detection boxes are located exactly over specific areas of faults. Although it does attain an acceptable accuracy level to detection, more computational resources for the frame

per second that has to be acquired compared to other options make it less scalable for industrial applications where speed is needed and hence not suitable for high-speed industrial inspection processes. The performance summary and real-time detection results from the FPGA implementation on the Xilinx PYNQ-Z2 board are shown in Figure 5. The output image clearly shows a mouse bite defect that was accurately detected under hardware-accelerated conditions, with a confidence score of 0.71. The high reliability of the suggested architecture is confirmed by the performance summary, which shows a latency of 3.54 seconds, an inference speed of 3.52 seconds, and perfect detection metrics with precision = 1.0, recall = 1.0, and MAE = 0.0. The FPGA fabric runs convolutional layers in parallel greatly reducing the computational burden and ensuring energy-efficient operation. For real-time industrial PCB inspection, it performs inference faster and more reliably than the CPU, ensuring accurate micro and macro flaws detection with low hardware costs as well as power consumption.

TABLE 1. PERFORMANCE MEASUREMENT OF YOLOv8N IN CPU & FPGA

PARAMETERS	YOLOv8n with CPU in PCB Defect Detection	YOLOv8n with FPGA in PCB Defect Detection
Precision	95.7%	98.5%
Recall	95.7%	98.5%
Mae	4.3%	1.5%
Inference	0.39 s	0.15 s
Latency	0.16 s	0.2 s

Table 1 depicts the performance measurement and comparison between YOLOv8n integrating with CPU and FPGA for defects identification in PCB defects. This achieves higher precision and recall of 98.5% rather than 95.7% on CPU. MAE values reduced from 4.3% to 1.5% which enhance the stability and makes it reliable one. The inference time measures about 0.15 seconds on FPGA and 0.39 on CPU. The convolutional operation increases its capacity with increases in latency about 0.2 seconds as it transfers the data from ARM and selects over CPU with less power and fast response. This supports the high-speed PCB inspection system in manufacturing sectors.

The evaluation of PCB defects is performed in the Xilinx PYNQ-Z2 board supported with ARM cortex A9 and FPGA fabric for hardware acceleration. The proposed methodology is provided in Fig. 1 and Fig. 2 depicts the defects of the PCB like open circuits, short, mouse bites and spurious copper traces. These defect samples were collected from standard PCB image datasets and used to train and validate the lightweight YOLOv8n model. The hardware platform setup is shown in Fig. 3, where the PYNQ-Z2 board is connected to the host system for configuration and testing. The trained YOLOv8n model is quantized and deployed to the FPGA fabric, enabling the convolutional layers to execute in parallel to reduce latency. Fig. 4. shows the CPU-based output image, where defect detection is performed via software-only inference. In comparison, Fig. 5. shows the output image generated by the PYNQ-Z2 FPGA, where the same defects are detected in real time with higher efficiency and reduced power

consumption. This reduces latency which shows the proposed system is applicable for providing real-time, high-throughput checking and highly desirable for industrial PCB manufacturing lines. This output confirms the proposed FPGA accelerated design improves processing speed and also increases scalability, effective for solution intelligent and low-cost and energy-efficient solutions for industrial environments for defect detections in PCB.

CONCLUSION

The real-time defects identification in PCB system runs on the Xilinx PYNQ-Z2 FPGA platform. It uses the lightweight YOLOv8n model, which has been optimized through quantization and pruning for efficient use in embedded settings. The FPGA design shows significant performance improvements. The precision and recall is all about 98.5% and duration of model to process the input is about 0.15 seconds, and a latency of 0.2 seconds. This surpasses CPU-based systems in both accuracy and speed. By using less power, FPGA is used to detect the fault in the parallel way. This system emerges to both tiny and large defects in PCB. This efficient method helps to detect the defects in automatic way with limited hardware. It is also reliable in defect identification system which can be used in the Industrial environments.

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